



TECHNISCHE
UNIVERSITÄT
DARMSTADT



Project lab

PRAKTIKUM ZUR INTELLIGENTEN ROBOTERMANIPULATION

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20-00-1170-PR

Summer Semester 2026

Praktikum zur intelligenten Roboteromanipulation: Part II - 20-00-1170-pr

- **Welcome to the Project Lab!**
- **Goal:** gain in-depth experience on a scientific project in AI & Robotic manipulation
- **How:** work by yourself or in a team of two students in a project that re-implements a recent research paper in the field. A list of papers is provided for you too choose from.
- **Projects:** each project paper comes from selected research topics connected to our research interests.

Projects have different requirements, depending on the paper associated to it. Some require implementation in the **real world** (on our robots in the lab), others only re-implementation in **simulation** with additional experimentation.

Note: students undergo a selection process to be allocated to a project. Apply for a project as soon as possible, by emailing the supervisors with a cover letter, CV and transcript of records. **The deadline for application is May 4th.**

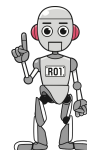


General information

- **Timeline:** April 24, 2026 — September 30th, 2026
- **Examination:**
 - Write a scientific report in **blogpost format** describing the work of the project
(deadline on Monday September 14th 2026, at 23:59 CET)
 - Hold a 15-minute presentation in class
(scheduled on Monday Sept 21st & Tuesday Sept 22nd – time TBA)
 - **Grading** : weighted average computed as **30% supervisor, 40% report, 30% presentation**
- **Contacts and Teaching assistants:**
 - Jiayun Li (*jiayun.li@tu-darmstadt.de*)
 - Prof. Georgia Chalvatzaki (*georgia.chalvatzaki@tu-darmstadt.de*)



Only for important matters





Projects



- You will be supervised by one/two of our PhD students.
- **Supervision:** supervision can be independently discussed with each phd student. In general, you can probably expect a meeting every two weeks, or feedbacks via message. They will guide you in studying the necessary material, and follow the right path to achieve the project's goals.
- **Implementation:** either in **simulation**, or in the **real world**. It depends on the specific project.
- **Goal:** in-depth dive into a recent research paper in the topic of AI for robot manipulation. It will give you a good idea of what it takes to fully understand a paper and reproduce its results.
- **Expected outcome:**
 - you feel more comfortable and have better critical thinking skills when reading cutting-edge research papers.
 - You learn about some open challenges in literature and how to solve them.
 - You learn what it takes to accomplish publication-level code implementations and results.



Projects

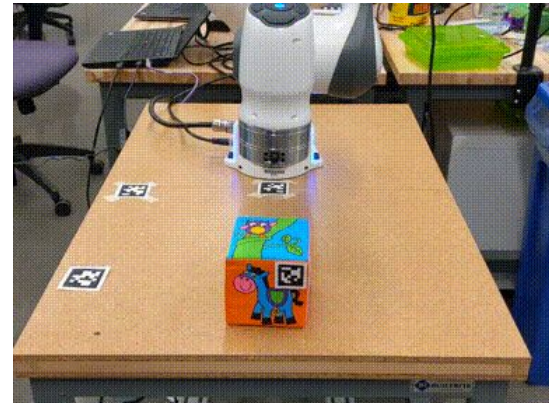


- **Expected outcome:**

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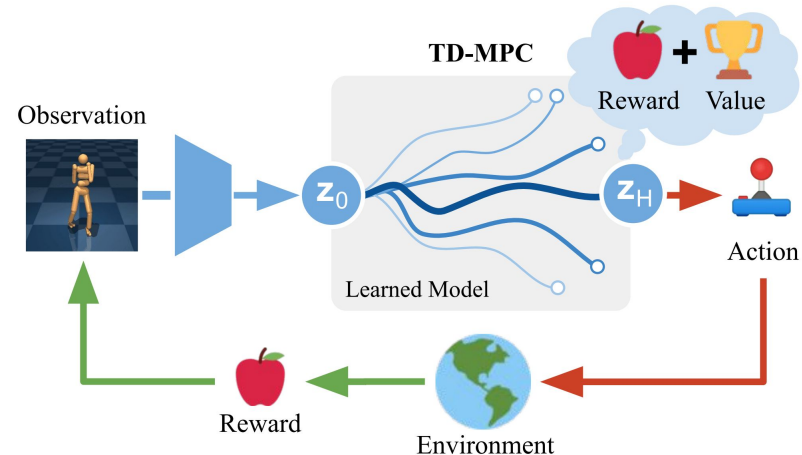
Robust Path Integral Control / Estimation for contact-rich Manipulation under Parameter Uncertainty

- This paper investigates the integration of model **predictive path integral control** , **online system identification** , and **object tracking** to achieve robust closed-loop performance, with the ultimate aim of bridging the notorious sim-to-real gap.
- **Resources:** [paper](#), [website](#), [code](#)
- **Goals / Expectations:**
 - Study and re-implement the method in simulation
 - Use the existing Path integral control pipeline and implement the robustness strategy and parameter update laws.
 - Integrate into the in-hand manipulation & tracking.
- **Requirements:**
 -  **Python** (advanced)
 -  **Robotics** & **Control** (basics)
 -  **Passion** (Advanced)
- **Contact to apply:** Jiayun Li (jiayun.li@tu-darmstadt.de)



TD-M(PC)2: Improving Temporal Difference MPC Through Policy Constraint

- We aim to explore the current **frontier** of the TD-MPC to further increase the **sample efficiency** and training wall-clock time. TD-MPC learns the reward, **latent dynamics**, actor and critic jointly.
- **Resources:** [paper](#), [website](#), [code](#)
- **Goals / Expectations:**
 - Study the roadmap of TD-MPC family and re-implement the method on the Orca hand scenario in simulation.
 - Study and explore different baselines and get a report of the performance impression. We will explore together and propose a future direction.
- **Requirements:**
 -  **Python and robotic simulation** (advanced)
 -  **RL** (theory and practical knowledge with deep RL)
 -  **Passion** (Advanced)
- **Contact to apply:** Jiayun Li (jiayun.li@tu-darmstadt.de)



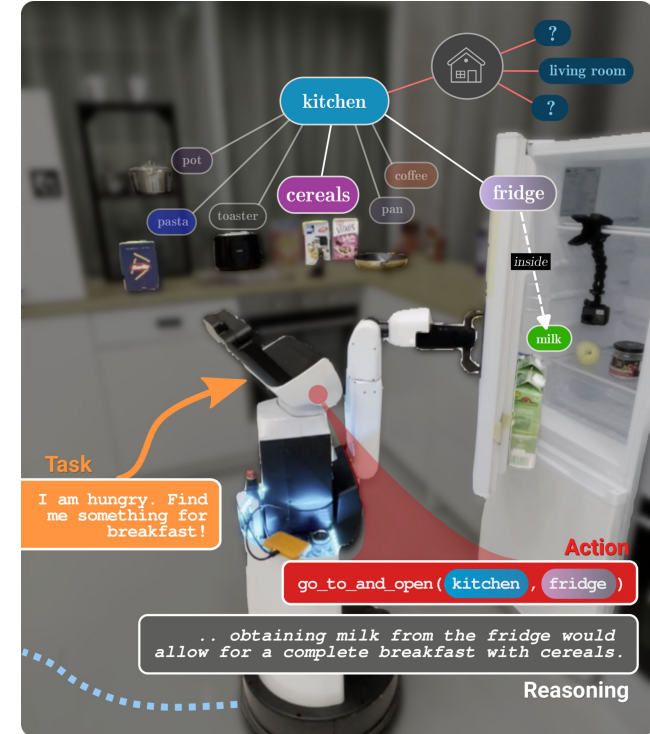
Memory representations for Robotic Task Planning

- This paper uses a VLM to evaluate action outcomes and self-reflect on failures, storing reflections in a short-term memory for within-task adaptation, that are then distilled into a long-term memory for retrieval-augmented generation of robot plans.
- **Resources:** [paper](#), [website](#), [code](#)
- **Goals / Expectations:**
 - Study and re-implement the method on our robot
 - Explore extensions for speeding up the system with local VLMs and incorporating other memory representations such as Ontology-based experiences.
- **Requirements:**
 -  **Python** (advanced)
 -  **Robotics** (basic experience with ROS)
 -  **LLMs** (practical knowledge of LLMs/VLMs, not mandatory)
 -  **Enthusiasm** (advanced)
- **Contact to apply:** Vignesh Prasad (vignesh.prasad@tu-darmstadt.de)



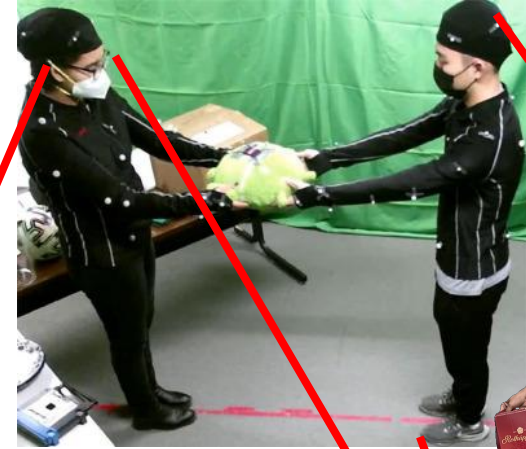
Scene-Graph guided Planning for Open-vocabulary Mobile Manipulation

- This paper uses Scene Graphs in high-level LLM reasoning for breaking down tasks in large-scale Long-Horizon Planning in Object Search.
- **Resources:** [paper](#), [website](#), [code](#)
- **Goals / Expectations:**
 - Study and re-implement the method in Simulation
 - Test the limits of the systems especially in recovering from mistakes and generating a dataset of such mistake recoveries.
- **Requirements:**
 -  **Python** (advanced)
 -   **LLMs** (some practical knowledge of using LLMs/VLMs)
 -  **Passionate** (important)
- **Contact to apply:** Vignesh Prasad (vignesh.prasad@tu-darmstadt.de)



Dataset of bimanual human-to-human object handovers

- Collecting a dataset of Bimanual Handovers using Egocentric and Exocentric RGBD cameras and annotating hand-object interactions.
- Resources:** [paper](#), [website](#)
- Goals / Expectations:**
 - Re-implement the data collection with our lab hardware, such as RGB-D cameras, Meta Aria Glasses
 - Incorporate gaze, hand and skeleton tracking using existing libraries.
 - Explore object detection and tracking from egocentric/exicentric cameras
- Requirements:**
 -  **Python** (advanced)
 -  **Robotics** (some basic experience with ROS)
 -  **Eagerness** (mandatory)
- Contact to apply:** Vignesh Prasad (vignesh.prasad@tu-darmstadt.de)



Audio for Contact-Rich Manipulation

- Those papers leverage audio as a sensing modality for contact-rich manipulation, focusing on (a) pretraining from internet-scale data and (b) training from in-the-wild data.
- Resources:** [paper: Hearing Touch](#), [paper: ManiWAV](#)
- Goals / Expectations:**
 - Extensive literature review for audio in manipulation (sensor setup, datasets, encoders).
 - Configuring and validating an audio sensor in ROS 2.
 - Feature extraction of collected audio(-vision) data and optional policy learning.

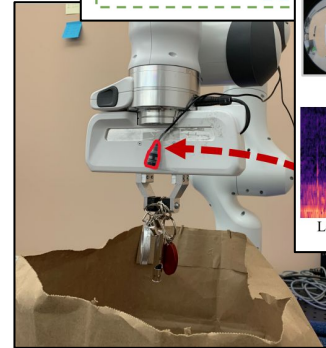
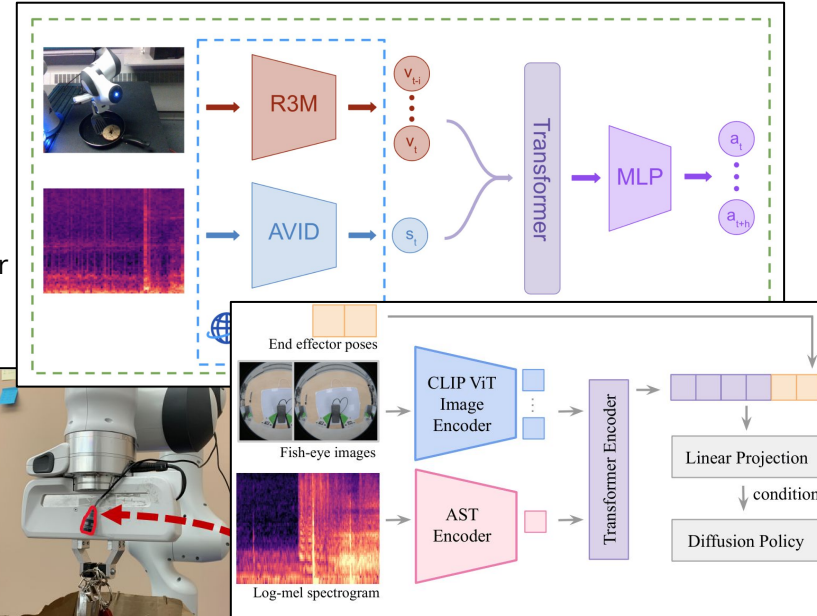
Requirements:



Python (advanced)

Robotics (some basic experience with ROS)

Motivation



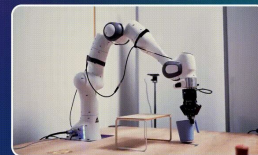
- Contact to apply:** Rickmer Krohn (rickmer.krohn@tu-darmstadt.de)

Beyond Isotropic Gaussian Regularization in LeWorldModel

- Replacing LeWorldModel (base code available) fixed isotropic Gaussian latent regularization with a structure-aware prior that adapts to environment complexity.
- **Resources:** [paper](#), [website](#)
- **Goals / Expectations:**
 - Reproduce the LeWM baseline on one or two small benchmark environments
 - Implement one structured latent regularization variant (e.g. anisotropic Gaussian, mixture-of-Gaussians, or sparsity-inducing prior)
 - Evaluate improvements in prediction quality, planning success, latent embedding structure, and training stability
- **Requirements:**
 - **Python** (experienced)
 - **Deep Learning / PyTorch** (understanding of loss functions)
 - World Models (willingness to learn)
- **Contact to apply:** Kumar Manas (kumar.manas@tu-darmstadt.de)



Start



Goal

The model receives a single image of the goal
(For pick-and-place, it receives two subgoals)

*<https://ai.meta.com/blog/v-jepa-2-world-model-benchmarks/>



Stressing the Limits of SE(3)-Equivariant Diffusion Policies

- When we grasp a rotated object, our hand should rotate accordingly. SE(3)-equivariant diffusion policies encode this symmetry as an inductive bias. But real robots face joint limits, collisions, and kinematic constraints.

👉 When does the symmetry prior truly help — and when does it break and hurt performance?

- **Resources:** [paper](#), [website](#), [code](#)

- **Goals / Expectations:**

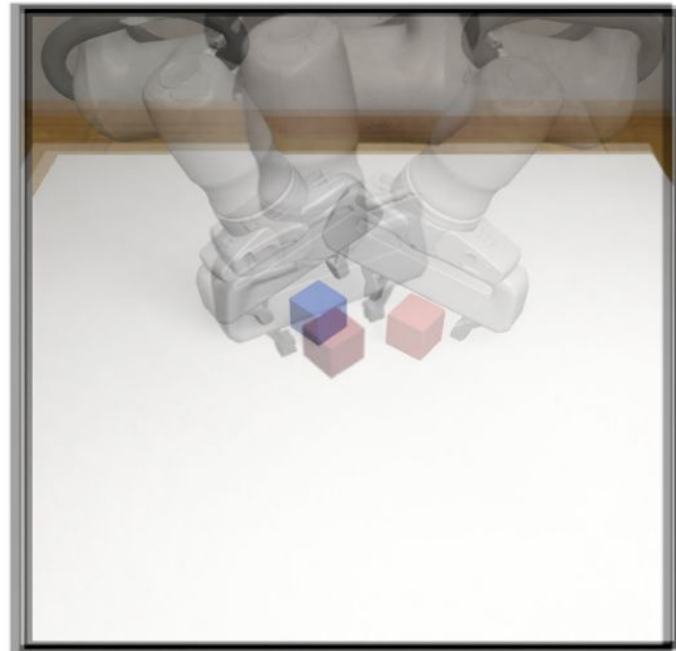
- Study and re-implement equivariant diffusion policy architectures
- Analyze symmetry assumptions and identify when they break
- Contribute to a benchmark that tests the limits of these policies in (mobile) manipulation

- **Requirements:**

- Background in robotics / robot learning
- Experience in Python and PyTorch
- *Bonus:* Experience in simulations, imitation learning
- *Bonus:* Equivariant learning, differential geometry

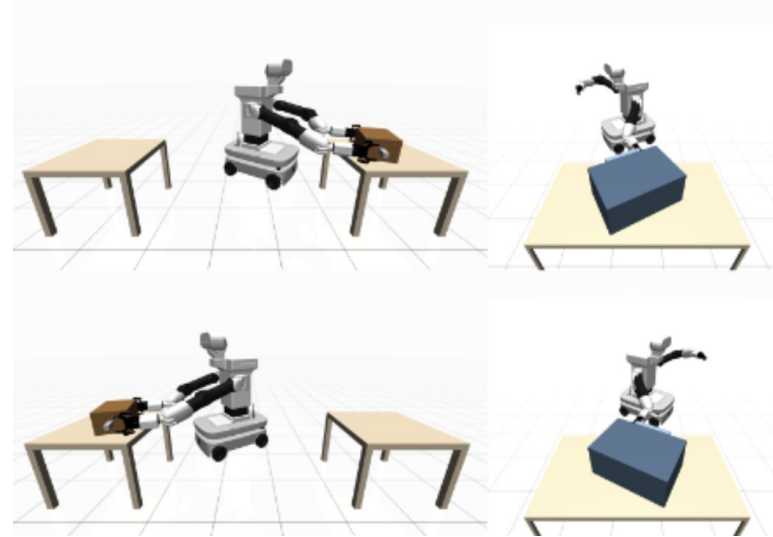
- **The topic is very suitable for a subsequent master thesis :)**

- **Contact to apply:** Max Siebenborn (max.siebenborn@tu-darmstadt.de)



VLAs for (Bimanual) Mobile Manipulation with Tiago

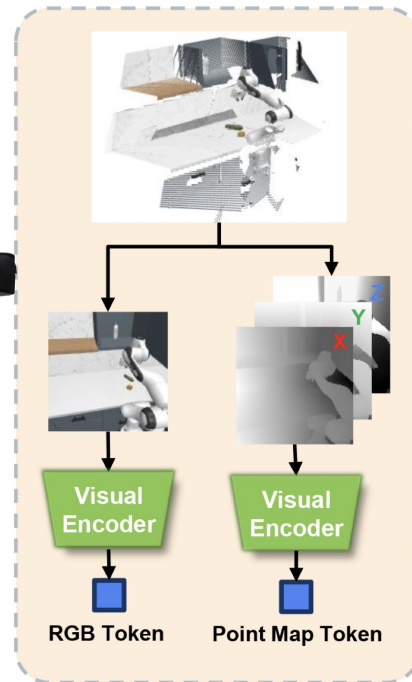
- Explore how Pi0.5 (and subsequent) models can be used on the Tiago++ mobile manipulator
- **Resources:** [paper](#), [website](#), [code](#)
- **Goals / Expectations:**
 - Integrate Pi0.5 (and potentially other VLAs) into our MuJoCo simulation suite and benchmark their performance
 - Benchmark their performance in mobile manipulation with > 18 DoFs
- **Requirements:**
 - Python, PyTorch / JAX
 - Ideally: experience with MuJoCo (or other simulators)
 - Ideally: experience in imitation learning / robot learning



- **Contact to apply:** Max Siebenborn (max.siebenborn@tu-darmstadt.de)

PointMap Policies for Mobile Manipulation with Tiago

- We will explore point maps as an observation modality for imitation learning in (bimanual) Mobile Manipulation with Tiago. Is it an effective & efficient modality for encoding visual & depth perception for Mobile Manipulation?
- **Resources:** [paper](#), [website](#)
- **Goals / Expectations:**
 - Integrate point map perception into our training and inference pipeline for IL in mobile manipulation
 - Benchmark pointmaps as an observation modality, comparing it against point clouds, rgbs, and states
- **Requirements:**
 - Python, ROS
 - Ideally: background in computer vision, robot learning
 - Wants to work with the real world :)
- **Contact to apply:** Sophie Lueth (sophie.lueth@tu-darmstadt.de)
Max Siebenborn (max.siebenborn@tu-darmstadt.de)



Convex Decompositions with Guarantees

- Convex decomposition is essential for computational geometry and robotic applications. The goal is to design novel algorithms that can do convex decomposition with guarantees from raw point clouds.

- Resources:** [paper1](#), [website](#), [code](#), [paper2](#)

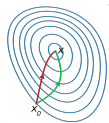
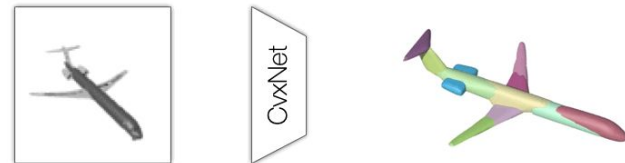
- Goals / Expectations:**

- Study and familiarize with existing convex decomposition methods – learning-based and optimization-based
 - Propose novel algorithms that can handle various constraints (keep number of convex sets as small as possible, keep the volume of convex sets as tight as possible)

- Requirements:**

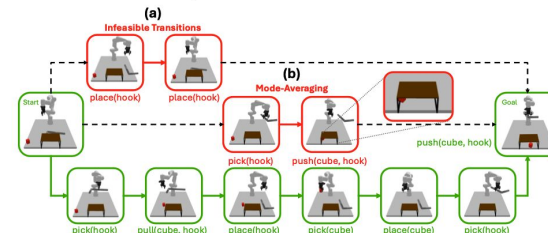
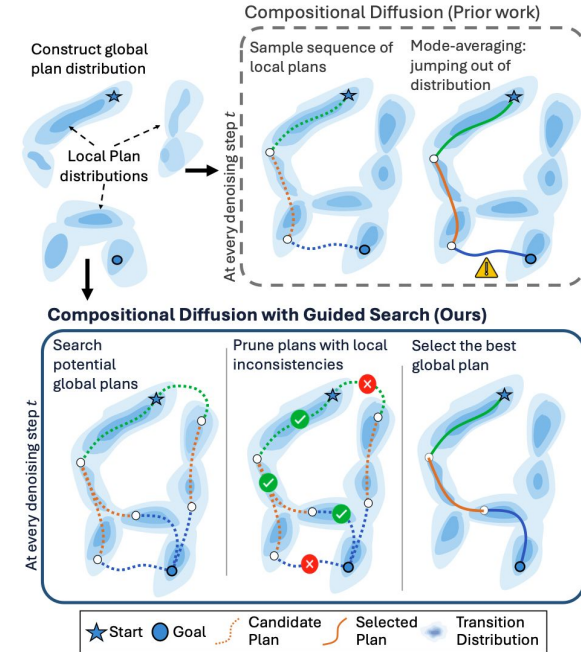
- Python** (advanced)
 - Optimization** (experienced), **deep learning** (experienced)
 - Julia** (willing to learn)
- suitable to extend as a Master Thesis**

- Contact to apply:** Yuezhe Zhang (yuezhe.zhang@tu-darmstadt.de)



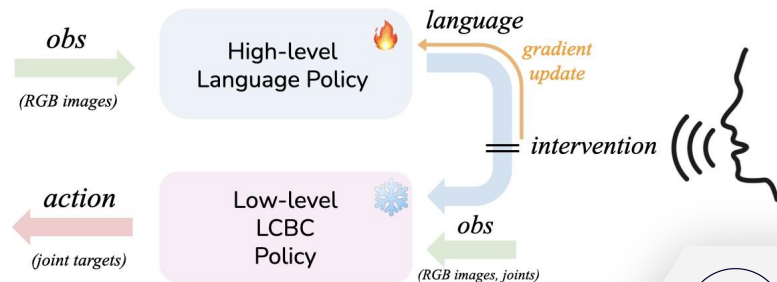
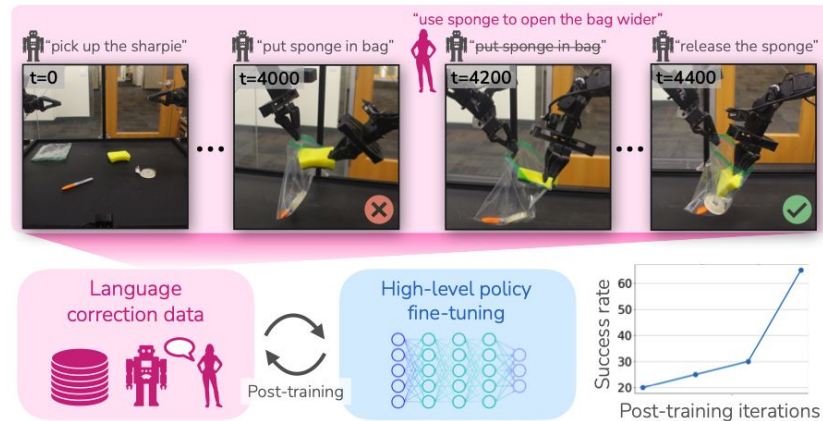
Compositional Diffusion with Guided Search for Long Horizon Planning

- The paper proposes a search based denoising process for diffusion.
- This prevents unfortunate side-effects like infeasible transitions and mode averaging.
- Resources:** [paper](#), [website](#), [code](#), [closely-related](#)
- Goals / Expectations:**
 - Recreate the sim robot TAMP experiments.
 - Using the above knowledge implement the system on a real world franka-panda robot.
- Requirements:**
 - Deep Learning (Pytorch) **[Advanced]** 
 - Mathematics related to Machine Learning **[Advanced]** $\sqrt{x}$
 - Robotics Knowledge (Python/ROS)  **[Upto IROBMAN course level]**
 - Software Engineering. (Python) **[Basic]** 
- Contact to apply:**
 Franziska Herbert (franziska.herbert@tu-darmstadt.de),
 Sohan Rudra (sohan.rudra@tu-darmstadt.de)



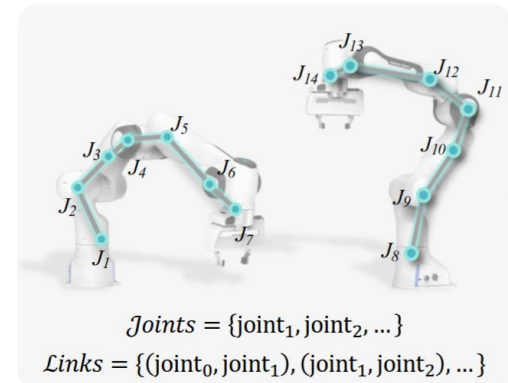
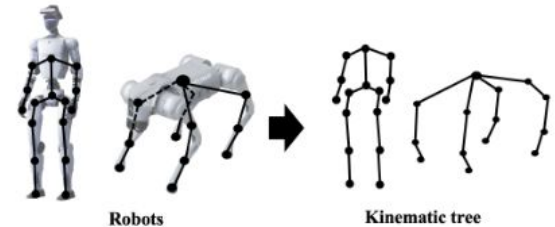
Yell at Your Robot

- The paper proposes an on the fly correction mechanism for high level policies solving long horizon tasks.
- This increases chance of success for long horizon tasks as it also continuously improves the High Level Policy.
- **Resources:** [paper](#), [website](#), [code](#)
- **Goals / Expectations:**
 - Study the method in detail and setup the data collection pipeline for our lab setups.
 - Implement the setup on a franka-panda robot.
- **Requirements:**
 - Deep Learning (Pytorch) **[Advanced]** 
 - Robotics Knowledge (Python/ROS) **[Upto IROBMAN course level]** 
 - Software Engineering. (Python) **[Advanced]** 
- **Contact to apply:**
 Franziska Herbert (franziska.herbert@tu-darmstadt.de),
 Sohan Rudra (sohan.rudra@tu-darmstadt.de)



Graph-Based Encoders for Robotic Structured Representations

- Investigate if robotic body structure improves learning
- The paper proposes a structured representation of the robotic structure and uses a dynamics-informed message passing algorithm for extracting relevant embeddings
- **Resources:** [primary paper](#), [related paper 1](#), [related paper 2](#)
- **Goals / Expectations:**
 - Reimplement the encoder and dynamics-informed message passing algorithms presented in the paper
 - Extend the representations to an **imitation learning** setting in simulation, e.g. Robocasa
 - Compare different graph encoders from related papers
- **Requirements:**
 - Deep Learning (Pytorch, Python) **[Advanced]** 
 - Prior knowledge on RL and Imitation Learning 
 - Experience with Simulators (Mujoco) is a plus
- **Contact to apply:**
Franziska Herbert (franziska.herbert@tu-darmstadt.de)



Efficient VLA Training with Object-Agent-centric Tokenization

- Vision-Language-Action (VLA) models are powerful but computationally expensive. This project adapts the OpenVLA model into Oat-VLA by introducing an inductive bias towards scene objects and the agent's visual information, drastically reducing visual tokens and converging 2x faster.
- **Resources:** [paper](#) , [OpenVLA](#)
- **Goals / Expectations:**
 - Study the Oat-VLA architecture and re-implement it starting from the OpenVLA base model.
 - Test and benchmark the implementation in simulation.
 - Deploy the model on a real robot in the lab to solve pick-and-place manipulation tasks.
- **Requirements:**
 - **Python and Deep Learning (PyTorch)** (advanced)
 - **Robotics** (basic experience with Simulators)
 - **LLMs/VLMs** (practical knowledge)
- **Contact to apply:** Ali Younes (ali.younes@tu-darmstadt.de)





What now?

Choose a project and apply for it by directly emailing the supervisors.

Deadline for application is May 4th, 2026.



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